

Simple command:

```
bhupathi@DESKTOP-2L0MACQ:~$ cd 2520090027_OSSP
bhupathi@DESKTOP-2L0MACQ:~/2520090027_OSSP$ cd Skill
bhupathi@DESKTOP-2L0MACQ:~/2520090027_OSSP/Skill$ mkdir Skill-4.c
bhupathi@DESKTOP-2L0MACQ:~/2520090027_OSSP/Skill$ cd Skill-4.c
bhupathi@DESKTOP-2L0MACQ:~/2520090027_OSSP/Skill/Skill-4.c$ gedit Skill-4.c
bhupathi@DESKTOP-2L0MACQ:~/2520090027_OSSP/Skill/Skill-4.c$ gcc Skill-4.c -o skill-4
bhupathi@DESKTOP-2L0MACQ:~/2520090027_OSSP/Skill/Skill-4.c$ ./skill-4
```

```
=====
      LINUX COMMAND LEXER AND PARSER
=====
```

Enter a Linux command:

> ls -l

```
===== TOKEN STREAM =====
```

Token 1: WORD -> ls

Token 2: WORD -> -l

Token 3: END -> END

```
=====
```

Token stream is valid.

```
===== PARSE / EXECUTION TREE =====
```

Command 1

Arguments: [ls] [-l]

```
=====
```

Pipe: `ls -l | grep txt`

```
bhupathi@DESKTOP-2L0MACQ:~/2520090027_OSSP/Skill/Skill-4.c$ ./skill-4
```

```
=====
      LINUX COMMAND LEXER AND PARSER
=====
```

Enter a Linux command:

> `ls -l | grep txt`

```
===== TOKEN STREAM =====
```

Token 1: WORD -> `ls`

Token 2: WORD -> `-l`

Token 3: PIPE -> `|`

Token 4: WORD -> `grep`

Token 5: WORD -> `txt`

Token 6: END -> END

```
=====
Token stream is valid.
```

```
===== PARSE / EXECUTION TREE =====
```

Command 1

Arguments: [`ls`] [`-l`]

|

V

Command 2

Arguments: [`grep`] [`txt`]

```
=====
```

Redirection: `ls > output.txt`

```
bhupathi@DESKTOP-2L0MACQ:~/2520090027_OSSP/Skill/Skill-4.c$ ./skill-4
```

```
=====
LINUX COMMAND LEXER AND PARSER
=====
```

```
Enter a Linux command:
```

```
> ls > output.txt
```

```
===== TOKEN STREAM =====
```

```
Token 1: WORD      -> ls
```

```
Token 2: OUTPUT    -> >
```

```
Token 3: WORD      -> output.txt
```

```
Token 4: END       -> END
```

```
=====
Token stream is valid.
```

```
===== PARSE / EXECUTION TREE =====
```

```
Command 1
```

```
Arguments: [ls]
```

```
Output    : output.txt
```

```
=====
```

Syntax error: ls |

```
bhupathi@DESKTOP-2L0MACQ:~/2520090027_OSSP/Skill/Skill-4.c$ ./skill-4
=====
      LINUX COMMAND LEXER AND PARSER
=====

Enter a Linux command:
> ls |

===== TOKEN STREAM =====
Token 1: WORD      -> ls
Token 2: PIPE      -> |
Token 3: END       -> END
=====
Syntax Error: Command cannot end with '|'.

```

Empty command: >

```
bhupathi@DESKTOP-2L0MACQ:~/2520090027_OSSP/Skill/Skill-4.c$ ./skill-4
=====
      LINUX COMMAND LEXER AND PARSER
=====

Enter a Linux command:
> >

===== TOKEN STREAM =====
Token 1: OUTPUT    -> >
Token 2: END       -> END
=====
Syntax Error: Redirection requires a filename.
```